WE CAN ACADEMY

Artificial Intelligence Course

END OF MODULE PROJECT

Months 1 – 2 | Group Assignment | 60 Marks

Group Name		Date Issued	01 / 04 / 2026
Group Members		Due Date	28 / 04 / 2026

Project Report: AI CV-Reviewer.

1. Cover Page.

- **Group Name**: Elite Group AI
- **Members' Adm no.s** : 4721, 2776 , 5198 ,1337 and 5114
- **GitHub Link**: https://github.com/Pmachariah-012/wca-ai-tool-elitegroup
- **Tool Name**: CV-Reviewer
- **Date**: 26/04/2026

2. Problem Statement.

Job seekers often face high rejection rates from automated screening systems because their CVs are not professionally optimized. While career coaching is effective, it is often too expensive for **university students, recent graduates**, **and unemployed youth** who are just starting their professional journeys. Furthermore, **mid-career professionals** looking to pivot and **individuals from** **underserved communities** often lack access to the tools needed to compete fairly.

Our tool, **CV-Reviewer AI**, provides immediate, expert-level feedback to help these candidates in the Kenyan job market improve their professional presentation. By removing the financial barrier to quality career advice, we empower **diverse talent pools** to increase their chances of success and secure the opportunities they deserve.

3. Tool Description

CV-Reviewer AI is a Python-based virtual recruitment assistant. The user interacts with the program by providing their professional experience as a string within the script. The tool transmits this data via the **OpenAI API** to the **gpt-4** model, which analyzes the text and returns a structured critique highlighting strengths and areas for improvement.

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4. AI Instruction Design (R-T-C-C-O)

The intelligence of the tool is guided by a highly specific prompt designed using the **R-T-C-C-O** framework to ensure structured output:

- **Role:** Senior Human Resource Manager and Expert Career Coach with 15 years of experience in the Kenyan corporate and tech industry.
- **Task:** Provide a high-quality critique identifying strengths, highlighting gaps, and suggesting industry-standard keywords.
- **Context:** The user is a university student or recent graduate in Kenya (e.g., from JKUAT) applying for entry-level positions.
- **Constraint:** Provide exactly 3 strengths and 3 weaknesses in a professional tone; ignore personal contact info; return response **ONLY as a valid JSON object**.
- **Output:** A structured JSON object containing `analysis_metadata`, `strengths`, `weaknesses`, an `action_plan`, and an `overall_score_out_of_10`.

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5. Technical Overview

The tool is built using **Python** and the **Requests** library to interface with Google's generative AI ecosystem.

- **API Configuration:** It connects to the **Google Gemini API** via the following endpoint: `https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash-latest:generateContent`.

- **Authorization:** Unlike standard Bearer Tokens, this integration uses an **API Key** passed directly as a query parameter in the URL string.
- **Logic:** The script captures user input via `user_cv`, formats it into the `CV_REVIEW_PROMPT` template, and sends a **POST** request with a JSON payload.
- **Parsing:** The tool navigates a complex JSON response body (`response.json()['candidates'][0]['content']['parts'][0]['text']`) to extract the AI's feedback.
- **Variables:** We used clear variable naming such as cv_text, ai_response, and prompt for readability.

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6. Challenges & Solutions

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- **Challenge 1: Handling API Responses.**
  - **Solution:** We implemented a conditional check (`if response.status_code == 200`) to ensure the program only attempts to parse data if the request was successful, otherwise printing a clear error message.
- **Challenge 2: Prompt Management.**
  - **Solution:** To keep the code clean and readable, we moved the complex R-T-C-C-O prompt into a separate `prompts.py` file and imported it into the main script.

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7. Ethics Reflection

Privacy: The script processes data in real-time and does not store user CV text in a database or log file after the session ends.

Bias: We acknowledge that AI feedback is based on general software engineering standards and may not account for every specific local industry variation

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8. Conclusion & Future Improvements

CV-Reviewer successfully demonstrates how local university students can leverage the **Gemini 1.5 Flash model** to receive professional career coaching. While the current version requires users to paste text directly into the console, future iterations will include:

- **Direct File Support:** The ability to upload `.pdf` or `.docx` files directly.
- **Job Matching:** A feature to compare the CV against a specific job description for a "compatibility score".
- **Security:** Moving the API Key from a hardcoded string to an environment variable (`.env`) for better security practices.

9.Appendix

Full Source Code:

main.py > ...

```python
import requests
from prompts import CV_REVIEW_PROMPT

API_KEY = "AIzaSyAfkgo2DT1NUNzd6AFPByUNli6Shqn2vNU"
URL = f"https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-flash-latest:generateContent?

def main():
    user_cv = input("Paste CV and hit Enter: ")

    # Send request to Gemini
    payload = {"contents": [{"parts": [{"text": CV_REVIEW_PROMPT.format(cv_text=user_cv)}]}]}
    response = requests.post(URL, json=payload)

    # Print result
    if response.status_code == 200:
        print(response.json()['candidates'][0]['content']['parts'][0]['text'])
    else:
        print("Error:", response.text)

if __name__ == "__main__":
    main()
```

prompts.py > ...

```python
# prompts.py

# ================================================================
# PROJECT: AI-Powered CV Reviewer
# FRAMEWORK: R-T-C-C-O (Role, Task, Context, Constraints, Output)
# ================================================================

CV_REVIEW_PROMPT = f"""
ROLE:
You are a Senior Human Resources Manager and Expert Career Coach with 15 years of experience
in the Kenyan corporate landscape and tech industry.

TASK:
Review the provided CV text. Your goal is to provide a high-quality critique that helps
the candidate stand out to recruiters by identifying strengths, highlighting gaps,
and suggesting industry-standard keywords.

CONTEXT:
The user is a university student or recent graduate in Kenya (likely from JKUAT or similar institutions)
applying for internships or entry-level positions in a competitive market.

CONSTRAINTS:
1. Provide exactly 3 specific strengths and 3 actionable weaknesses.
2. The tone must be professional, constructive, and empowering.
3. Ignore or redact personal contact information (phone numbers, physical addresses).
4. You MUST return the response ONLY as a valid JSON object. Do not include any conversational text before c

OUTPUT FORMAT (JSON):
{{
  "analysis_metadata": {{
```

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```

OUTPUT FORMAT (JSON):
{{
  "analysis_metadata": {{
    "candidate_level": "Entry-Level",
    "market_relevance": "High"
  }},
  "strengths": [
    "First strength here",
    "Second strength here",
    "Third strength here"
  ],
  "weaknesses": [
    "First area for improvement",
    "Second area for improvement",
    "Third area for improvement"
  ],
  "action_plan": [
    "Specific step to take now",
    "Specific keyword to add"
  ],
  "overall_score_out_of_10": 0
}}

CV TEXT TO ANALYZE:
{{cv_text}}.
"""
```

README.md > abc # wca-ai-tool-elitegroup

```markdown
# wca-ai-tool-elitegroup
AI-powered CV reviewer tool using Python and API.
# WCA AI CV Reviewer Tool

##  Project Overview
The **CV Reviewer Tool** is a Python-based application designed to provide professional, automated feedback on resumes. By leveraging AI API integration and advanced prompt engineering, the tool analyzes CV content to help users improve their professional presentation.

This project was developed as part of the **Elite Group** collaboration, focusing on technical implementation, ethical AI use, and structured data output.

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##  Features
* **R-T-C-C-O Prompting:** Utilizes a robust prompt framework (Role, Task, Context, Constraints, Output) for high-accuracy reviews.
* **JSON Data Formatting:** All reviews are parsed into JSON for clean, structured, and readable output.
* **CLI Interface:** Simple command-line operations for user input and processing.
* **Personalized Feedback:** Tailors suggestions based on specific user career goals and background.

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##  Project Structure
* `prompts.py`: The engine of the tool, containing the R-T-C-C-O logic.
* `main.py`: Handles the API communication, user input, and script execution.
* `README.md`: Documentation and project guide.

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```

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SUBMISSION CHECKLIST .

- ☐ Python script (.py file) runs without errors
- ☐ API key is working and requests are being made
- ☐ R-T-C-C-O prompt is documented in the report
- ☐ Written report covers all 9 required sections
- ☐ Code has comments throughout GitHub repo created with correct naming format.
- ☐ Every group member has at least 3 commits on GitHub
- ☐ README.md is complete on the GitHub repo
- ☐ GitHub link is included on the report cover page
- ☐ Group has rehearsed the live demo at least once
- ☐ All group members know and can explain the code
- ☐ Each student has submitted individually on Google Classroom
- ☐ Personal reflection paragraph is included in your submission
- ☐ Report includes full source code in the appendix